Banking Chatbot

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Introduction

- This is Banking full-stack project.
- My task was to add a chatbot for question-answering support.
- I added a chatbot for internal support; it is configured to answer questions based only on our database knowledge and does not pull information from external sources.

Tech Stack used :

- **Frontend:** React (Vite), JavaScript/JSX,CSS.
- **Browser APIs:** Fetch API, localStorage
- **Backend:** Python 3.11, FastAPI, Uvicorn
- **Config/validation:** pydantic, pydantic-settings
- **RAG/LLM:** LangChain, OpenAI API
- **Retrieval/indexing:** FAISS , PyPDFLoader / pypdf
- **Data source:** PDF knowledge base only

Requirements :

requirements.txt

backend > requirements.txt

```
fastapi==0.115.12
uvicorn[standard]==0.34.0
pydantic==2.11.3
pydantic-settings==2.10.1
python-dotenv==1.0.1
langchain==0.3.23
langchain-openai==0.3.13
langchain-community==0.3.21
openai==1.109.1
faiss-cpu==1.13.2
pypdf==5.4.0
numpy==2.2.3
```

Frontend : Chatbot UI

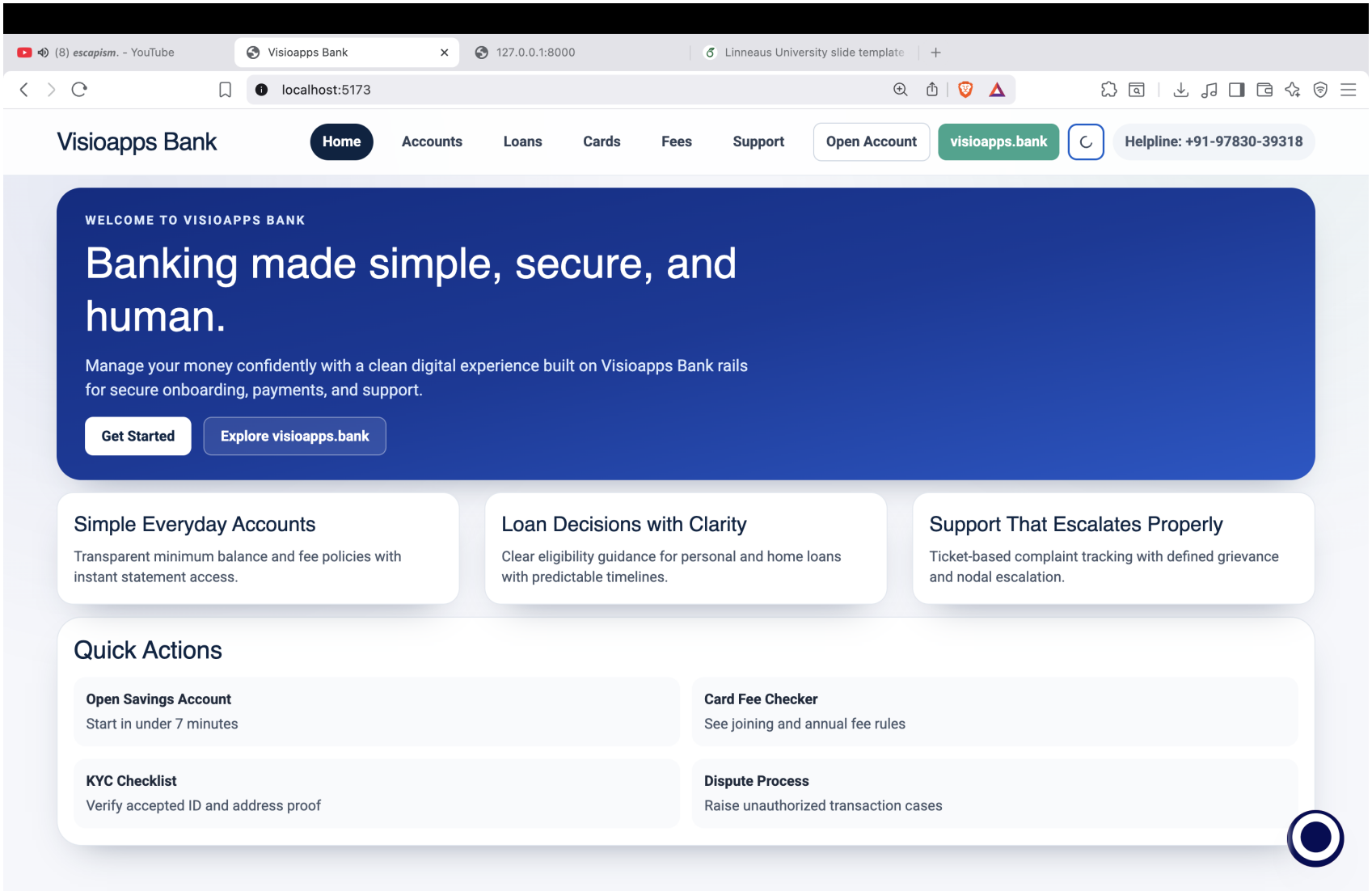

Figure: User Interface

Chatbot UI (Multilingual support)

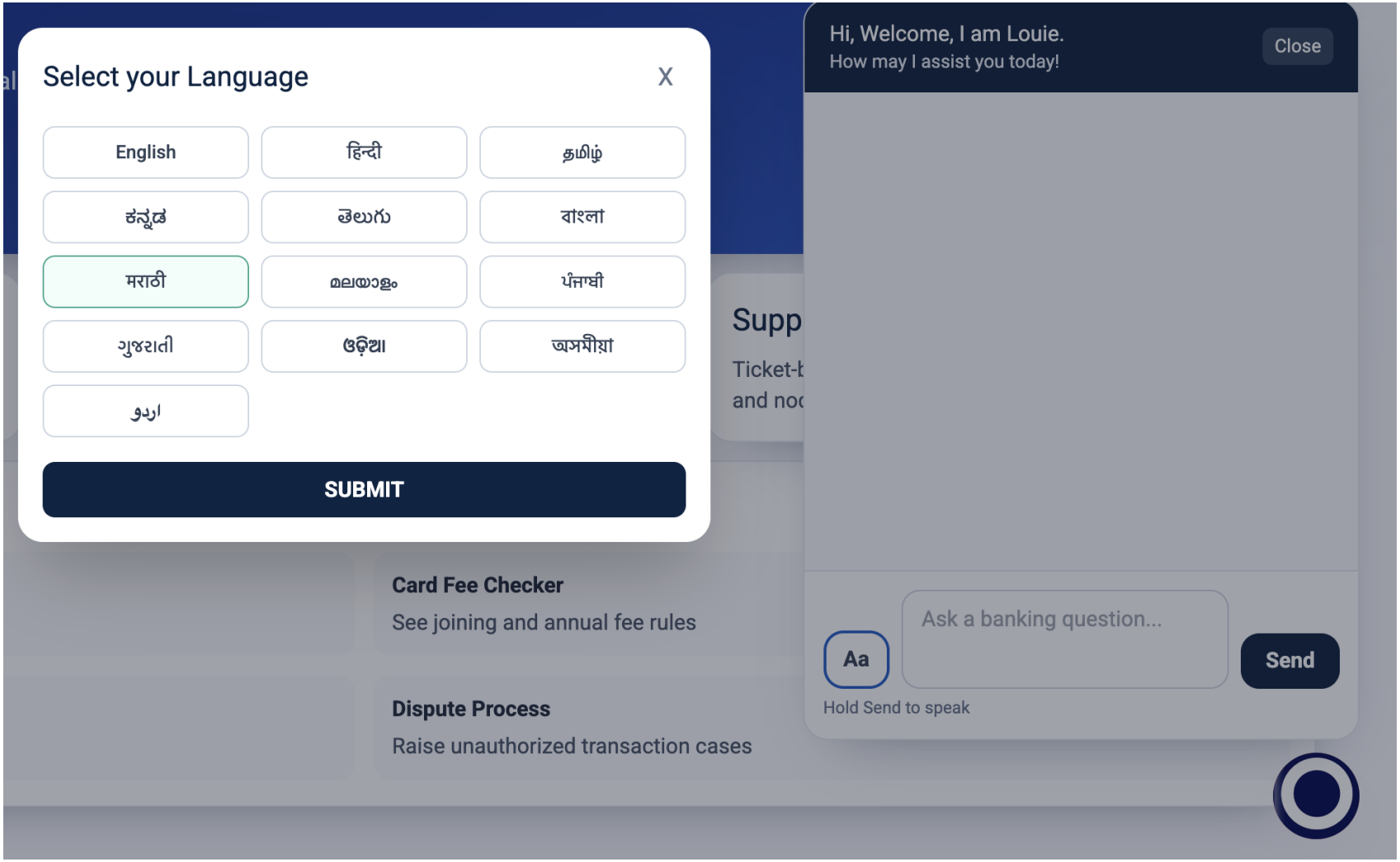

Figure: User Interface

Chatbot QA

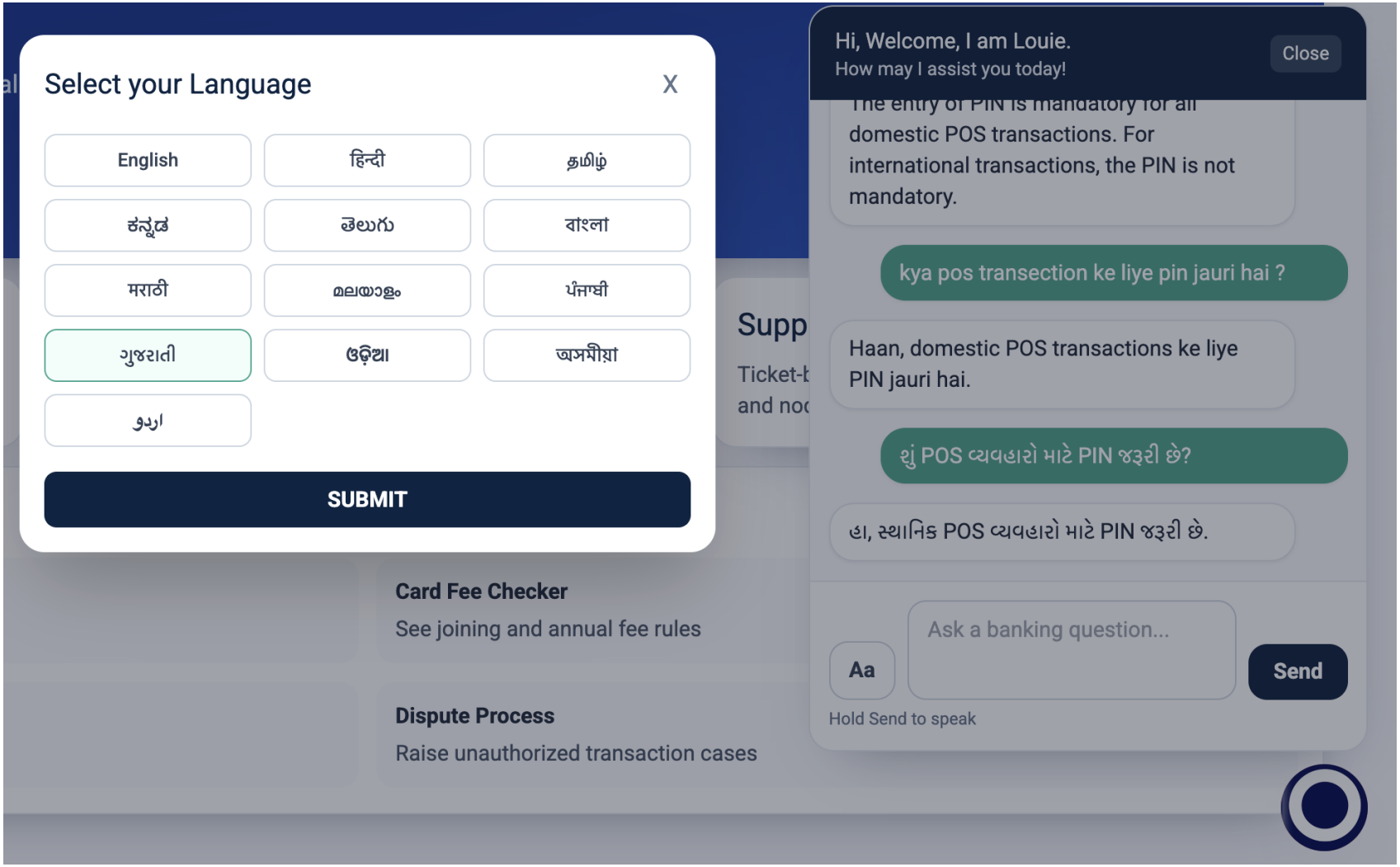

Figure: Chatbot QA User Interface

Chatbot QA

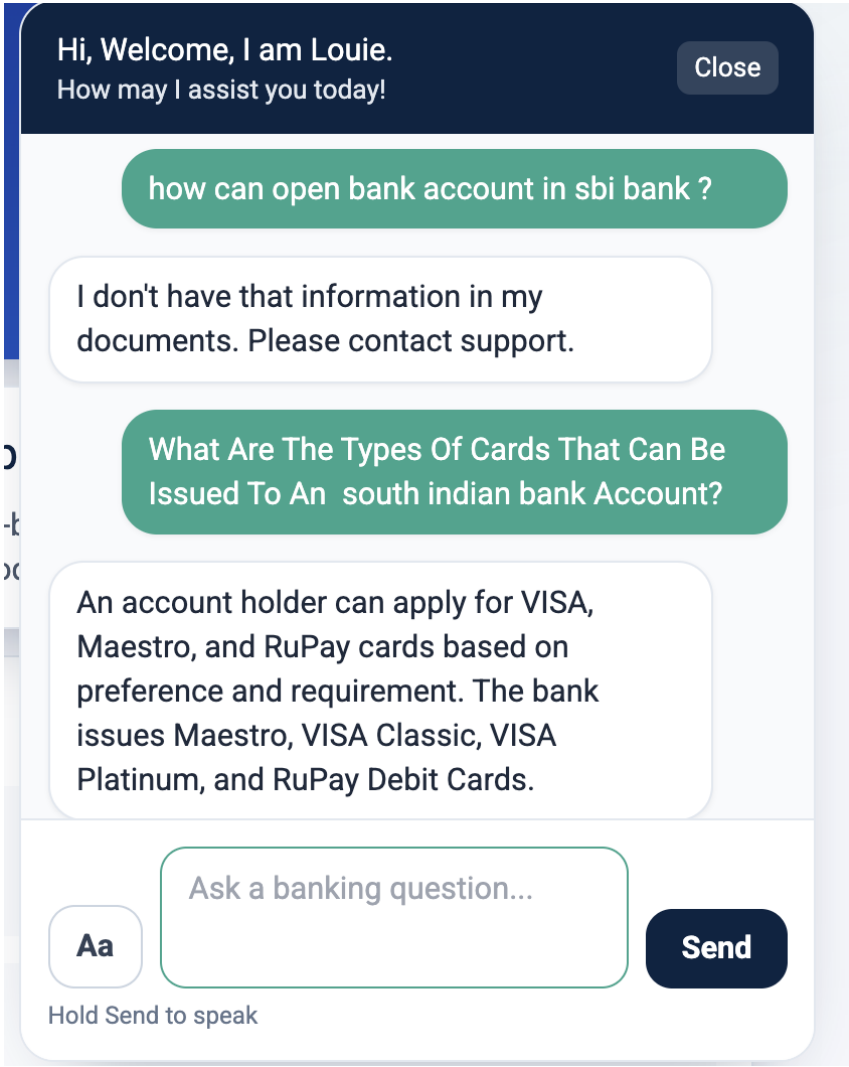

Figure: Chatbot QA User Interface

Parameters :

```
OPENAI_API_KEY=your_openai_api_key_here
FRONTEND_ORIGIN=http://localhost:5173
OPENAI_CHAT_MODEL=gpt-4o-mini
OPENAI_EMBEDDING_MODEL=text-embedding-3-large
TOP_K=4
SIMILARITY_THRESHOLD=0.46
CHUNK_SIZE=1100
CHUNK_OVERLAP=180
GENERATION_MAX_TOKENS=320
MAX_MESSAGE_LENGTH=1000
RATE_LIMIT_REQUESTS=20
RATE_LIMIT_WINDOW_SECONDS=60
```

GitHub

Repository

github.com/avii2/Chatbot_Support_Banking_webSite

Thank You!

Questions?

DM me on LinkedIn: anilkbarupal